Lesson 7 Data Dictionary

This dictionary describes the SQLite warehouse in `clickstream_warehouse.db` and the clickstream document shape used by the project ETL.

SQLite Warehouse

The data types below are the types declared in the current lesson database. Row counts are from the inspected database snapshot.

Table	Column(s)	Data type	Business definition
`dim_courses`	`code_module`, `code_presentation`	TEXT	Course module and presentation identifiers; together form the logical course-presentation key.
`dim_courses`	`module_presentation_length`	TEXT	Length of the course presentation in days. The current database stores this value as TEXT.
`dim_assessments`	`id_assessment`	INTEGER	Unique assessment identifier and logical key.
`dim_assessments`	`code_module`, `code_presentation`	TEXT	Course presentation associated with the assessment.
`dim_assessments`	`assessment_type`	TEXT	Assessment category, such as TMA, CMA, or Exam.
`dim_assessments`	`date`	INTEGER	Assessment day offset from the course presentation start; may be unavailable.
`dim_assessments`	`weight`	REAL	Assessment contribution to the course grade, usually expressed as a percentage.

Table	Column(s)	Data type	Business definition
dim_vles	id_site	INTEGER	Unique identifier for a Virtual Learning Environment (VLE) resource; logical key.
dim_vles	code_module, code_presentation	TEXT	Course presentation containing the VLE resource.
dim_vles	activity_type	TEXT	Type of online learning resource or activity.
dim_vles	week_from, week_to	INTEGER	Optional first and last course weeks when the resource is available.
dim_students	id_student	INTEGER	Anonymized student identifier.
dim_students	code_module, code_presentation	TEXT	Course presentation associated with the student record. Together with id_student, forms the logical key.
dim_students	gender, region, highest_education, imd_band, age_band, disability	TEXT	Recorded demographic and background categories for the student.
dim_students	num_of_prev_attempts	INTEGER	Number of earlier attempts at the course.
dim_students	studied_credits	INTEGER	Number of credits the student is studying.
dim_students	final_result	TEXT	Final course outcome, such as Pass, Fail, Distinction, or Withdrawn.
fact_registration	id_student	INTEGER	Anonymized student identifier; joins to the student-course record.
fact_registration	code_module, code_presentation	TEXT	Registered course presentation. Combined with

Table	Column(s)	Data type	Business definition
			`id_student`, forms the logical key.
`fact_registration`	`date_registration`, `date_unregistration`	INTEGER	Registration and unregistration day offsets relative to the course presentation; either may be unavailable.
`fact_assessment_results`	`id_assessment`, `id_student`	INTEGER	Assessment and student identifiers; together form the logical result key.
`fact_assessment_results`	`date_submitted`	INTEGER	Submission day offset relative to the course presentation.
`fact_assessment_results`	`is_banked`	INTEGER	Indicator for a result carried over from an earlier attempt (0 or 1).
`fact_assessment_results`	`score`	REAL	Assessment score on a 0-100 scale; may be unavailable.
`agg_bucket_engagement`	`id_student`, `id_site`	INTEGER	Student and VLE resource identifiers for an engagement aggregate. No unique key is declared.
`agg_bucket_engagement`	`total_clicks`, `active_days`, `first_day`, `last_day`, `early_clicks`	INTEGER	Aggregated click count, number of active days, first and last activity offsets, and clicks in the early-engagement period.

Logical Keys and Relationships

Table	Logical key	Relationship
`dim_courses`	`code_module` + `code_presentation`	Parent course-presentation record for course dimensions and related facts.

Table	Logical key	Relationship
dim_assessments	id_assessment	code_module + code_presentation identify its course presentation.
dim_vles	id_site	code_module + code_presentation identify its course presentation.
dim_students	code_module + code_presentation + id_student	Student record for a specific course presentation.
fact_registration	code_module + code_presentation + id_student	Joins to the corresponding student-course record.
fact_assessment_results	id_assessment + id_student	Joins to the assessment and student dimensions.
agg_bucket_engagement	Not declared	id_student and id_site relate to student and VLE records.

The keys and relationships above describe the warehouse model. The current SQLite schema does not declare foreign-key constraints, and its generated tables do not declare NOT NULL or CHECK constraints. ETL validation or governance rules should not be described as database-enforced unless those rules have been applied and verified.

MongoDB Clickstream

The ETL stores clickstream events in VLE_Logs.clickstream_events as compact documents grouped by student and VLE site. Date and click arrays are positionally aligned.

Collection	Field	BSON type	Business definition
clickstream_events	_id	Integer	Generated bucket identifier based on the student and VLE site.
clickstream_events	id_site	Integer	VLE resource identifier; relates to dim_vles.id_site.
clickstream_events	d	Array of integers	Activity day offsets for the bucket.
clickstream_events	n	Array of integers	Click counts corresponding by position to the values in d.

Snapshot Counts

Table	Rows
dim_courses	22

Table	Rows
dim_assessments	195
dim_vles	6,364
dim_students	32,593
fact_registration	32,593
fact_assessment_results	171,047
agg_bucket_engagement	0